2024－2025春高数下期末（回忆版）

1. Multiple Choice Questions: (only one correct answer for each of the following questions.)

(1) The divergence of the vector field $F=\langle xy^2, ye^z, x\ln(1+z^2)\rangle$ at the point $(1,1,0)$ is

(A) -1 (B) 0 (C) 1 (D) 2

(2) Let $f(x,y)$ be a function defined on the entire plane such that $f_x(x,y)<0$ and $f_y(x,y)>0$. Which of the following conditions guarantees that $f(x_1,y_1)<f(x_2,y_2)$?

(A) $x_1<x_2$, $y_1<y_2$. (B) $x_1<x_2$, $y_1>y_2$.

(C) $x_1>x_2$, $y_1<y_2$. (D) $x_1>x_2$, $y_1>y_2$.

(3) $\int_{-1}^{0}\int_{-x}^{2-x^2}(1-xy)\,dydx+\int_{0}^{1}\int_{x}^{2-x^2}(1-xy)\,dydx=$

(A) $\frac{7}{3}$ (B) $\frac{7}{6}$ (C) $\frac{5}{3}$ (D) $\frac{5}{6}$

(4) $\lim\limits_{(x,y)\to(0,0)}\dfrac{\sin(x^4y^2)}{(x^4+y^2)^2}=$

(A) 0 (B) 1 (C) $\frac{1}{2}$ (D) The limit does not exist.

(5) Consider the following four properties of a function with two variables:

① $f(x,y)$ is continuous at point (x_0,y_0);

② The partial derivatives of $f(x,y)$ at point (x_0,y_0) are continuous;

③ $f(x,y)$ is differentiable at point (x_0,y_0);

④ The partial derivatives of $f(x,y)$ exist at point (x_0,y_0).

If "$P\Rightarrow Q$" denotes that property P implies property Q, which of the following relationships is correct?

(A) ②⇒③⇒① (B) ③⇒②⇒① (C) ③⇒④⇒① (D) ③⇒①⇒④

2. Fill in the blanks.

(1) The equation for the tangent plane at the point $(1,0,1)$ on the surface $z=x^2(1-\sin y)+y^2(1-\sin x)$ is _____.

(2) Let D be the region $\{(x,y)\mid x^2+y^2\le 1\}$, then $\iint_D(x^2+y^2)\,dxdy=$ _____.

(3) The maximum directional derivative of $f(x,y,z)=x^2e^y+\cos(y^2+z^2)$ at the point $(1,0,0)$ is _____.

(4) $\lim\limits_{x\to 0}\dfrac{e^{-x-\frac{1}{2}x^2}-1+2x-\sin x}{x^3}=$ _____.

(5) If $\dfrac{(x+ay)\,dx+y\,dy}{(x+y)^2}$ is the total differential for some function, then $a=$ _____.

3. A curve C defined by the parametric equations $\begin{cases}x=t-\sin t,\\ y=1-\cos t.\end{cases}$

(1) Find the equation of the tangent line to the curve C at the point where $t=\frac{\pi}{3}$.

(2) Find $\frac{d^2y}{dx^2}$.

4. Determine whether the following series converges absolutely, or converges conditionally, or diverges. Give reasons for your answer.

(1) $\sum_{n=1}^{\infty} \frac{(-1)^n n^3}{2^n}$.

(2) $\sum_{n=2}^{\infty} (-1)^{n+1} \frac{1}{n \ln n}$.

5. Find all the local extreme values of $f(x,y) = xe^{-\frac{x^2+y^2}{2}}$.

6. Find the centroid of the region $\{(x,y,z) \mid x^2+y^2 \leq z \leq 1\}$.

7. Find the outward flux of

$$F = (3xy^2 + e^{y^2}\cos(xy^2+z^2))\,i + (6x^2y - yz^2)\,j + (\frac{z^3}{3} + \ln(x^2+y^2))\,k.$$

across the boundary of the region $\{(x,y,z) \mid 0 \leq z \leq 1-(2x^2+y^2)\}$.

8. Calculate the line integral

$$\oint_C (\frac{\sin x}{1+x^2} - y)\,dx + (x - z + e^{y^2})\,dy + (x - y + \ln(1+2z^2))\,dz,$$

where the curve C is defined by $\begin{cases} x^2+y^2-2y=0 \\ x-y+z=2 \end{cases}$

and its orientation is counterclockwise when viewed from above.